

# Identification of perturbative ambiguity canceled against bion

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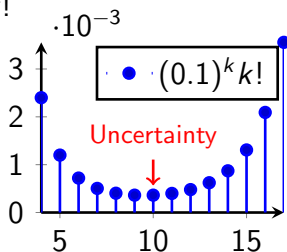
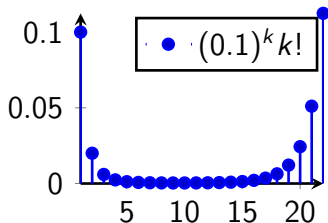
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- PLB **807** (2020) 135570 [arXiv:2003.04759 [hep-th]]

# Factorial growth of perturbation series

- In QM/QFT, perturbative expansions are divergent series
- Typically,  $F(\lambda) = \sum_{k=0}^{\infty} c_k \lambda^k$ ,  $c_k \sim k!$



## ① Proliferation of Feynman diagrams

Ground state energy

Zeeman effect  $\sim (-1)^k (2k)!$

Stark effect  $\sim (2k)!$

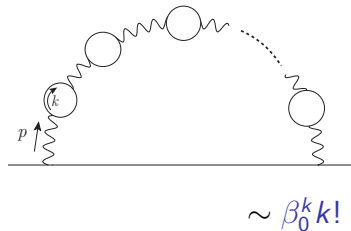
$V(\phi) \sim \phi^3 \sim \Gamma(k + 1/2)$

$V(\phi) \sim \phi^4 \sim (-1)^k \Gamma(k + 1/2)$

Double well  $\sim k!$

Periodic cosine  $\sim k!$

## ② Renormalon ['t Hooft '79]



# Borel resummation and perturbative ambiguity

- Borel resummation: summing divergent asymptotic series

$$f(g^2) \sim \sum_{k=0}^{\infty} f_k \left( \frac{g^2}{16\pi^2} \right)^{k+1} \quad \text{with } f_k \sim a^k k! \text{ as } k \rightarrow \infty$$

$\Downarrow$  Borel transform

$$B(u) \equiv \sum_{k=0}^{\infty} \frac{f_k}{k!} u^k = \frac{1}{1 - au} \quad (\text{Pole singularity at } u = 1/a).$$

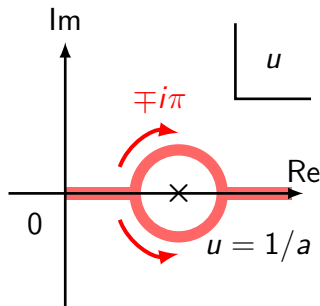
- The Borel sum is given by

$$f(g^2) \equiv \int_0^{\infty} du B(u) e^{-16\pi^2 u/g^2}.$$

- $a < 0$  (alternating series)  $\rightarrow$  convergent

- $a > 0 \rightarrow$  ill-defined due to the pole

$\Rightarrow$  Imaginary ambiguity  $\sim \pm e^{-16\pi^2/(ag^2)}$



# Resurgence theory and semi-classical picture

- Resurgence structure

$$\delta_{\text{PT}} \sim e^{-16\pi^2/(ag^2)} \overset{\text{Cancellation}}{\longleftrightarrow} \text{Nonperturbative effects}$$

① Instanton (one-instanton action  $S_I = 8\pi^2/g^2$ )

② Bion [Ünsal '07]: fractional instanton/anti-instanton pair

on  $\mathbb{R}^{d-1} \times S^1$  with  $\mathbb{Z}_N$ -twisted boundary conditions (BC)

|                           | PT                                                           | Semi-classical object                               |
|---------------------------|--------------------------------------------------------------|-----------------------------------------------------|
| $\mathbb{R}^4$            | $\delta_{\text{PFD}} \sim e^{-16\pi^2/g^2}$                  | $\delta_{\text{instanton}} \sim e^{-2S_I}$          |
| $\mathbb{R}^4$            | $\delta_{\text{renormalon}} \sim e^{-16\pi^2/(\beta_0 g^2)}$ |                                                     |
| $\mathbb{R}^3 \times S^1$ |                                                              | $\delta_{\text{bion}} \sim e^{-2S_B} = e^{-2S_I/N}$ |

(E.g.,  $\beta_0 = \frac{11}{3}N$  for  $SU(N)$  GT)

- Reading out nonperturbative effects from PT: Resurgence theory

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| $\mathbb{R}^4$            | $\delta_{\text{renormalon}} \sim e^{-16\pi^2/(\beta_0 g^2)}$ | ?                                                   |
| $\mathbb{R}^3 \times S^1$ | ?                                                            | $\delta_{\text{bion}} \sim e^{-2S_B} = e^{-2S_I/N}$ |

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- Reading out nonperturbative effects from PT: Resurgence theory
- What about QFT?
  - Conjecture: renormalon  $\leftrightarrow$  Bion [Argyres–Ünsal '12, ...]
  - But, no renormalon on  $\mathbb{R}^3 \times S^1 \rightarrow$  Conjecture [16pSA-4]

# Our work: Enhancement of PFD ambiguity

- What cancels bion ambiguities?
- Answer: **PFD-type ambiguity** (cf. [Anber–Sulejmanpasic '14])

$$\delta_{\text{PFD}}|_{\mathbb{R}^4} \sim e^{-16\pi^2/g^2} \stackrel{?}{\leftrightarrow} \delta_{\text{bion}}|_{\mathbb{R}^3 \times S^1} \sim e^{-2S_B} = e^{-2S_I/\textcolor{red}{N}}$$

- **Enhancement of PFD upon compactification** [O.M.–Takaura '20]

$$\text{PFD: } k! \rightarrow \textcolor{red}{N}^k k!$$

$$\delta_{\text{PFD}} \sim e^{-16\pi^2/g^2} \rightarrow \delta'_{\text{PFD}} \sim e^{-16\pi^2/(\textcolor{red}{N}g^2)}$$

- Keys (similar to Linde problem ['80] in finite-temperature QFT):
  - ①  $S^1$  compactification  $\rightarrow$  IR divergences
  - ② twisted BC  $\rightarrow$  twist angles as IR regulators
- Let us study the  $\mathbb{C}P^N$  model on  $\mathbb{R} \times S^1$ 
  - ▶ Bion calculus [Fujimori–Kamata–Misumi–Nitta–Sakai '18]

## 2-dimensional $\mathbb{C}P^N$ model with twisted BC

- 2D  $\mathbb{C}P^{N-1}$  model ( $A = 1, 2, \dots, N$ )

$$S = \frac{1}{g^2} \int d^2x \left[ \partial_\mu \bar{z}^A \partial_\mu z^A - j_\mu j_\mu + f(\bar{z}^A z^A - 1) \right]$$

where  $j_\mu = (1/2i) \bar{z}^A \overleftrightarrow{\partial}_\mu z^A$ ,  $f$  is an auxiliary field ( $\bar{z}^A z^A = 1$ )

► # of vacuum bubble diagrams  $\sim 4^k \Gamma(k + 1/2)$

- $\mathbb{Z}_N$ -twisted BC:

$$z^A(x_1, x_2 + 2\pi R) = e^{2\pi i m_A R} z^A(x_1, x_2), \quad m_A R = \begin{cases} A/N & \text{for } A \neq N \\ 0 & \text{for } A = N \end{cases}$$

- Amplitude of a diagram ( $F^{2k}$ :  $2k$ th-order polynomial,  $f_0 = \langle f \rangle$ )

$$\xrightarrow{p_2=0, A=1} \frac{V_2(g^2)^k}{(2\pi R)^{k+1}} \int \left( \prod_{i=1}^{k+1} \frac{dp_{i,1}}{2\pi} \right) \frac{F^{2k}(p_{i,1}, m_A)}{\prod_{i=1}^{2k} [q_{i,1}^2 + m_A^2 + f_0]}$$

IR divergence in massless limit  $m_A^2 + f_0 \rightarrow 0$  ( $\int d^2p \rightarrow \int dp$ )

# IR structure and enhancement mechanism

- $m_A^2 + f_0 = 1/(NR)^2 + f_0$  works as an IR regulator

- ▶  $m_A^2 \gg f_0$

$$\sim \frac{V_2}{R^2} \frac{1}{(m_A R)^{k-1}} \left( \frac{g^2}{4\pi} \right)^k \xrightarrow{A=1} \frac{V_2}{R^2} \frac{1}{N} \left( \frac{N g^2}{4\pi} \right)^k \quad \text{Enhancement!}$$

- ▶  $m_A^2 \ll f_0$

$$\sim \frac{V_2}{R^2} \sum_{\alpha \geq 0} \frac{(m_A R)^\alpha}{(f_0 R^2)^{(k+\alpha-1)/2}} \left( \frac{g^2}{4\pi} \right)^k \quad \text{Enhancement}$$

- Dependence on  $NR\Lambda$  ( $\Lambda$ : dynamical scale)

- ▶  $NR\Lambda \ll 1$  (Bion calculus is valid)

$$\sqrt{f_0} R \sim \frac{g^2}{4\pi} \Rightarrow [m_A^2 = \mathcal{O}(g^0)] \gg [f_0 = \mathcal{O}(g^4)] \quad \text{Enhancement!}$$

- ▶  $NR\Lambda \gg 1$  (Large  $N$ )

$$f_0 \sim \Lambda^2 \Rightarrow \frac{m_A^2}{f_0} = \frac{1}{(NR\Lambda)^2} \ll 1 \quad \text{Enhancement}$$

# Summary and discussion

- Resurgence structure on  $\mathbb{R}^{d-1} \times S^1$

|                           | PT                                              | Semi-classical object                      |
|---------------------------|-------------------------------------------------|--------------------------------------------|
| $\mathbb{R}^4$            | $\delta_{\text{PFD}} \sim e^{-16\pi^2/g^2}$     | $\delta_{\text{instanton}} \sim e^{-2S_I}$ |
| $\mathbb{R}^3 \times S^1$ | $\delta'_{\text{PFD}} \sim e^{-16\pi^2/(Ng^2)}$ | $\delta_{\text{bion}} \sim e^{-2S_I/N}$    |
| $\mathbb{R}^3 \times S^1$ | $\delta_{\text{renormalon}} = 0$ [16pSA-4]      | No                                         |

- Enhancement phenomenon is consistent with bion
- Borel singularities at  $m_A R = A/N = \underbrace{1/N}_{1\text{-bion}}, \underbrace{2/N}_{2\text{-bion}}, \dots$
- $N R \Lambda \gg 1$  [Ishikawa-O.M.-Shibata-Suzuki '20]  
vs  $N R \Lambda \ll 1$  [Fujimori-Kamata-Misumi-Nitta-Sakai '18]

$$\begin{array}{ccc}
 E_{\text{large } N} \sim N^{-1}(\Lambda R)^{-2} \delta \epsilon^2 & \Rightarrow f_0 = \Lambda^2 \text{ in denominator} \\
 \updownarrow & \downarrow (m_A^2 \text{ vs } f_0) \\
 \text{Im } E_{1\text{-bion}} \sim \pm N(\Lambda R)^2 \delta \epsilon^2 & \Leftarrow \text{enhancement of } N
 \end{array}$$

- Helpful in giving a unified understanding on resurgence